



SOEN 6841: SOFTWARE PROJECT MANAGEMENT

TOPIC: FINANCIAL LITERACY APP

Group 1

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PROBLEM IDENTIFICATION REPORT

1.1 Problem Statement

Investors have long sought reliable ways to preserve and grow their wealth, yet despite the numerous opportunities offered by financial markets, investment decisions are often influenced by human emotions, cognitive biases, and subjective judgment. Behavioural finance research, particularly by Nobel laureates like Richard Thaler and Daniel Kahneman, has shown that investors frequently make irrational decisions based on emotional reactions to market fluctuations. This often results in suboptimal investment choices, such as buying high and selling low, which can lead to significant financial losses. Studies have revealed that the average investor's returns lag the market by nearly 4% annually, largely due to these behavioural biases.

To overcome these challenges, a data-driven, emotion-free approach to stock investing is essential. Our project aims to develop an intelligent application that leverages concrete numerical data and advanced algorithms, such as High Quantitative Momentum (HQM) Analysis, to provide objective, strategic, and data-backed stock investment advice. By removing human intervention in the stock shortlisting process, our solution will empower investors with unbiased insights, ultimately leading to more optimized and rational decision-making.

This approach draws from successful strategies used by leading quantitative investment firms like Renaissance Technologies and Two Sigma, which have outperformed the market by utilising complex algorithms and large datasets, free from the emotional biases that typically distort investment decisions. By harnessing similar techniques, our application provides a powerful tool for investors seeking to make decisions grounded in empirical evidence rather than emotion, potentially narrowing the performance gap between average investors and market benchmarks.

1.2 Objective

The purpose of this project is to develop a highly efficient Stock Selection application that empowers users to make informed and data-driven investment decisions. The application will aid in selecting stocks and determining their quantity for trading, considering crucial factors like capital allocation, risk tolerance, and desired diversification.

- i. **Data-driven stock selection:** Utilize quantifiable financial data to guide stock choices, ensuring unbiased decision-making.

- ii. **Risk assessment:** Evaluate individual risk appetite to tailor stock recommendations that align with each user's comfort level.
- iii. **Diversification Strategy:** Offer strategies that help users diversify their investments for balanced portfolio growth.
- iv. **Minimized human bias:** Remove emotional and psychological influences from the decision-making process, leading to more objective outcomes.
- v. **Enhanced precision in investment choices:** Leverage advanced algorithms to increase the accuracy and effectiveness of stock selection.
- vi. **Portfolio optimization:** Enable users to create well-balanced portfolios based on their financial goals and risk profile.
- vii. **User confidence:** Ensure that the stock selection process is transparent, empowering users to make decisions with confidence.

1.3 Stakeholders

Stakeholders in a project are individuals or groups who have a vested interest, involvement, or influence over the project's development and outcomes. In the case of a Financial Literacy App, stakeholders play crucial roles in shaping its success. Below are the key stakeholders, along with their interests and responsibilities:

Users	• Stock investors who prioritize long-term investment strategies and sustainable wealth growth. • Investors who value data-driven decision-making, recognizing that quantifiable metrics provide more reliable investment insights than subjective intuition or gut feeling.
Developers	• Developers are responsible for building, maintaining, troubleshooting, and optimizing the App to ensure a secure, user-friendly, and data-driven investment experience.
Product Managers	• Oversee the app's development and strategy, ensuring it aligns with user needs, market trends, and business objectives. • Coordinate between stakeholders, including developers, investors, and analysts, to enhance functionality, resolve issues, and drive continuous improvements.
Marketing Team	• Promote the app to the target audience through digital marketing, social media, and strategic campaigns to drive user adoption and engagement. • Gather user feedback and market insights to refine marketing strategies and collaborate with the product team for continuous improvements and feature enhancements.
Investors	• Provide financial support and resources to fund the development, marketing, and expansion of the Financial Literacy App. • Offer strategic guidance and industry insights to ensure the app remains competitive, scalable, and aligned with market needs.
Regulatory Bodies	• Ensure compliance with financial regulations by monitoring the app's adherence to securities laws, investment advisory guidelines, and data privacy standards. • Protect investor interests by enforcing transparency, accuracy, and ethical practices in stock recommendations to prevent misinformation and market manipulation.
Competitors	• Traditional mutual funds and hedge funds serve as competitors, with mutual funds lacking flexibility in risk and diversification, while hedge funds have high capital entry barriers. • Automated trading platforms like Coinrule offer algorithmic trading but primarily focus on cryptocurrencies rather than comprehensive stock market investments.

Figure 1: List of Stakeholders and Relationships

1.4 Relevance to Software Solutions:

Investors often struggle with emotion-driven and biased decision-making, leading to suboptimal investment choices. Psychological factors such as fear, greed, and cognitive biases can cloud judgment, causing investors to overreact to market fluctuations or follow trends without proper analysis. This issue is further exacerbated for retail investors who may lack the time, expertise, or resources to conduct in-depth financial research.

To address this challenge, we propose a data-driven, automated Stock Selection application that removes emotional biases and provides objective, quantifiable investment insights. By utilizing High Quantitative Momentum (HQM) analysis and algorithmic models, the application will offer a structured and efficient approach to stock selection.

1.4.1 Key features and benefits:

- i. **Unbiased, rule-based investment decisions:** It eliminates emotional and subjective biases by relying on pre-defined, quantifiable investment models. Using HQM analysis, it identifies high-potential stocks based on momentum indicators, past performance, and risk-adjusted returns.
- ii. **Catering to both institutional and retail investors:** Institutional investors, including hedge funds, mutual funds, and private equity managers, benefit from improved capital allocation strategies and optimized portfolio diversification. Retail investors, on the other hand, can access advanced stock selection methods without requiring deep financial expertise.
- iii. **Customized investment strategies:** The system adapts to individual risk appetite and diversification preferences, ensuring tailored investment recommendations while providing data-driven insights to help investors construct well-balanced, risk-adjusted portfolios.
- iv. **Real-Time Data Analysis and Insights:** It continuously monitors market trends, company fundamentals, and stock momentum to refine investment choices, helping investors make informed decisions based on concrete financial data rather than speculation.
- v. **Scalable and Systematic Approach to Investing:** The solution ensures a structured decision-making process that can be scaled for both individual and institutional use, enhancing efficiency by automating stock selection and reducing the time spent on research and manual analysis.

By integrating these features, the Stock Selection application will provide investors with an intelligent, data-backed decision-support tool, enabling them to make rational, optimized, and emotion-free investment choices.

1.5 Scope

The Stock Selection Application is designed to provide a structured, data-driven approach to investing, leveraging High Quantitative Momentum (HQM) analysis to remove human bias and enhance decision-making. The scope of the application encompasses a wide range of functionalities, ensuring it caters to both individual retail investors and institutional portfolio managers seeking optimized investment strategies.

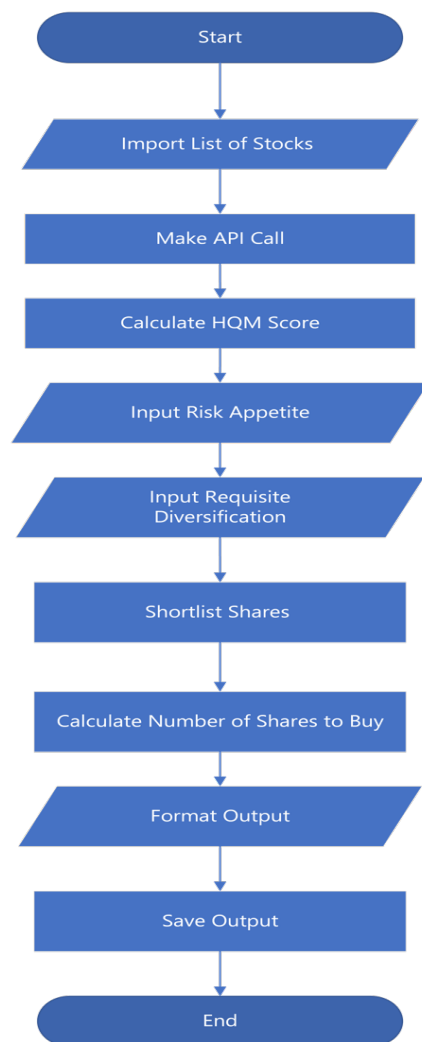


Figure 2: Flowchart showing proposed usage

1.5.1 Core Functionalities:

- i. **Automated Stock Shortlisting:** The application utilizes HQM analysis to systematically evaluate and rank stocks based on momentum trends, past performance, and risk-adjusted returns. By integrating real-time market data and financial analytics, it ensures that investors can make well-informed decisions without requiring extensive manual research. The shortlisting process is fully automated, reducing the need for subjective decision-making and enhancing efficiency in-stock selection.
- ii. **Risk Management and Capital Allocation:** Recognizing that investors have diverse risk appetites and capital constraints, the application enables users to define their investment profiles. The system dynamically adjusts stock recommendations based on the user's preferred risk exposure—whether conservative, moderate, or aggressive. This feature ensures that investors can balance their portfolio risk effectively while maximizing potential returns.
- iii. **User Profile Customization:** The application is designed to cater to a broad spectrum of investors, including retail investors, hedge funds, mutual fund managers, and private equity professionals. Users can customize their investment strategies based on individual financial goals, time horizons, and diversification preferences. By allowing investors to define specific parameters, such as minimum or maximum stock weightings and sector preferences, the app ensures a high degree of personalization.
- iv. **Long-Term Investment Data:** Staying ahead of market trends is essential for successful investing. The application continuously monitors market conditions, providing real-time updates on stock performance, industry trends, macroeconomic indicators, and risk factors. By leveraging this comprehensive data, the system analyzes historical patterns and current market conditions to devise a well-balanced stock portfolio tailored to the user's risk profile and diversification preferences. This ensures that investment decisions are data-driven, strategic, and aligned with long-term financial goals.
- v. **Intuitive User Interface and Accessibility:** A seamless user experience is central to the application's design. The interface is optimized for ease of use, with interactive tools that guide users through the investment process. Whether accessed via web or mobile, the application provides an intuitive and responsive layout that accommodates both novice and experienced investors. Additionally, the platform incorporates accessibility features, ensuring usability for individuals with different levels of financial expertise.

- vi. **Regulatory Compliance and Security:** Given the sensitive nature of financial data, the application is designed with stringent security measures, including encryption protocols and multi-factor authentication. Compliance with financial regulations, such as those established by the Investment Industry Regulatory Organization of Canada (IIROC) and the Canadian Securities Administrators (CSA), ensures that the application adheres to industry best practices.

1.5.2 Future Scope and Expansion:

Beyond its initial implementation, the application has the potential to evolve into a comprehensive investment ecosystem. Planned enhancements include:

- i. **Integration with Trading Platforms:** The ability to execute trades directly within the application by linking brokerage accounts.
- ii. **AI-Powered Investment Strategies:** Machine learning algorithms to predict market trends and automate rebalancing strategies.
- iii. **Multi-Asset Class Support:** Expansion to include ETFs, bonds, cryptocurrencies, and alternative investments.

By integrating these core functionalities and planned enhancements, the Stock Selection Application aims to become a powerful tool for investors seeking data-driven, risk-optimized, and systematic investment strategies.

MARKET ANALYSIS

2.1 Market Overview and Positioning

Understanding the financial landscape is crucial for the successful adoption and implementation of an investment-driven application. The Stock Selection Application aims to bridge the gap between data-driven investment strategies and accessibility for both retail and institutional investors. This market analysis evaluates the target audience, competitive landscape, and unique value propositions that set our solution apart.

The increasing adoption of quantitative investing and algorithmic stock selection highlights a shift from traditional, intuition-based investment approaches to more structured, data-backed decision-making. In Canada, as well as globally, investors are seeking automated, transparent, and efficient solutions that remove human bias from stock selection. By providing a scalable, systematic, and user-friendly platform, our application caters to a broad spectrum of investors—from experienced portfolio managers to individual retail investors looking for simplified yet effective investment strategies.

The following sections analyse the key demographics, user behaviours, and competitive landscape to establish the application’s market positioning and growth potential.

2.2 Target Audience for the Software Solution

The primary audience for this application includes individuals seeking long-term investment strategies but lacking the expertise or time to actively choose stocks to manage their portfolios. These users recognize that making investment decisions based on quantifiable data is a more reliable approach than relying on gut instincts or emotional biases. The project aims to make investing both accessible and intuitive, empowering users to make informed, data-driven decisions for their long-term financial goals.

2.2.1 Typical Users and Context:

The beauty of this solution is that it’s scalable and flexible. And can be adopted by individuals with a professional background in finance or a regular retail investor. Some of these could likely be:

- i. **Hedge Fund Managers:** Hedge fund managers can leverage this application to retrieve and analyze data, integrating it into their proprietary investment strategies to optimize portfolio performance and meet client needs efficiently.
- ii. **Mutual Fund Managers:** This tool enables mutual fund managers to allocate capital more effectively by systematically shortlisting stocks based on quantitative analysis, reducing reliance on traditional committee-driven investment decisions.
- iii. **Private Equity Managers for UNI (Ultra High Net Worth Individuals):** Similar to hedge fund managers, private equity professionals can utilize the application's data-driven insights to enhance their investment strategies, ensuring informed decision-making and improved asset allocation.
- iv. **Retail Investors:** Retail investors, who are the primary motivation behind this application, can benefit from a simplified, systematic approach to stock selection. Even individuals with minimal stock market knowledge can use the platform to make informed investment decisions, tailoring their portfolios based on their risk appetite and diversification preferences to achieve sustainable returns.

2.2.2 Context of Usage:

The platform serves as a decision-support tool, helping users apply quantitative investing strategies in real-world scenarios to shortlist stocks based on their risk and diversification chosen.

- i. **For Professional Investors (Hedge Fund Managers, Mutual Fund Managers, Private Equity Managers):**
 - a. Serves as a data-driven investment assistant, streamlining asset allocation and decision-making.
 - b. Reduces dependency on traditional committee-driven investment strategies by implementing a structured, algorithmic approach to stock selection.
 - c. Enhances capital allocation efficiency through automated weightage recommendations, optimizing portfolio diversification and risk management.
- ii. **For Retail Investors and Beginners:**
 - a. Eliminates emotional biases in decision-making, ensuring investment choices are based on quantifiable data.

- b. Assists users in constructing well-diversified portfolios without requiring advanced financial expertise.
- c. Provides a simulation feature that allows users to test various market conditions and evaluate systematic investing strategies in real-time.
- d. Promotes long-term wealth accumulation by encouraging disciplined, data-backed investment practices over short-term speculative trading.

2.2.3 Alignment with User Needs:

- i. **For Non-Experts:** Enables systematic investing with bare minimum market expertise.
- ii. **For Institutional Investors:** Enhances workflow and decision-making efficiency and improves capital allocation strategies.
- iii. **For Risk-Conscious Long-term Investors:** Reduces decision fatigue by following predefined, rule-based investment models.

2.3 Demographic and Psychographic Characteristics of the Target Audience

2.3.1 Demographic Characteristics:

The demographics can be visualized in the charts below

Category	Description
Age	22–60 years old (Young professionals, institutional investors, and retirees looking for automated investment solutions).
Gender	Broad appeal across male and female investors.
Income	Wide range: Middle-income earners (\$40,000–\$100,000) seeking stable growth. High-income earners (\$100,000+) who prefer automation in wealth management.
Location	Global adoption, particularly in markets where systematic investing is gaining traction: U.S., U.K., Canada, Australia, India, and Europe.

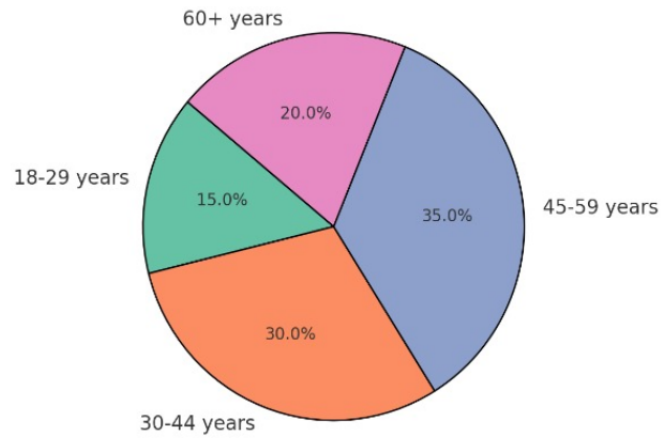


Figure 3: Age Distribution of Long-Term Stock Investors

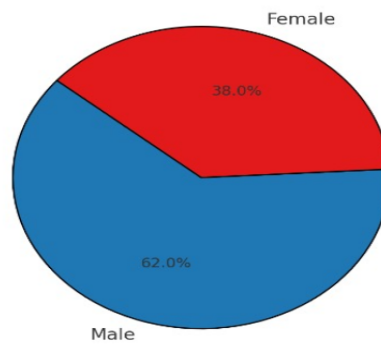


Figure 4: Gender Distribution of Long-Term Stock Investors

2.3.2 Psychographic Characteristics:

The psychographics can be visualized in the charts below

Category	Description
Investment Approach	Prefers rule-based investing strategies (e.g., Quantitative Momentum, Passive Indexing) over speculative trading.
Risk Tolerance	Moderate to low-risk investors who value stability, automation, and long-term wealth creation.

Technology Adoption	Comfortable using data-driven investment tools but may not have deep coding or quantitative finance expertise.
Lifestyle and Behaviour	- Seeks financial independence and passive wealth building. - Prefers automated investment strategies over hands-on stock picking. - Wants reliable, transparent decision-making without emotional bias.

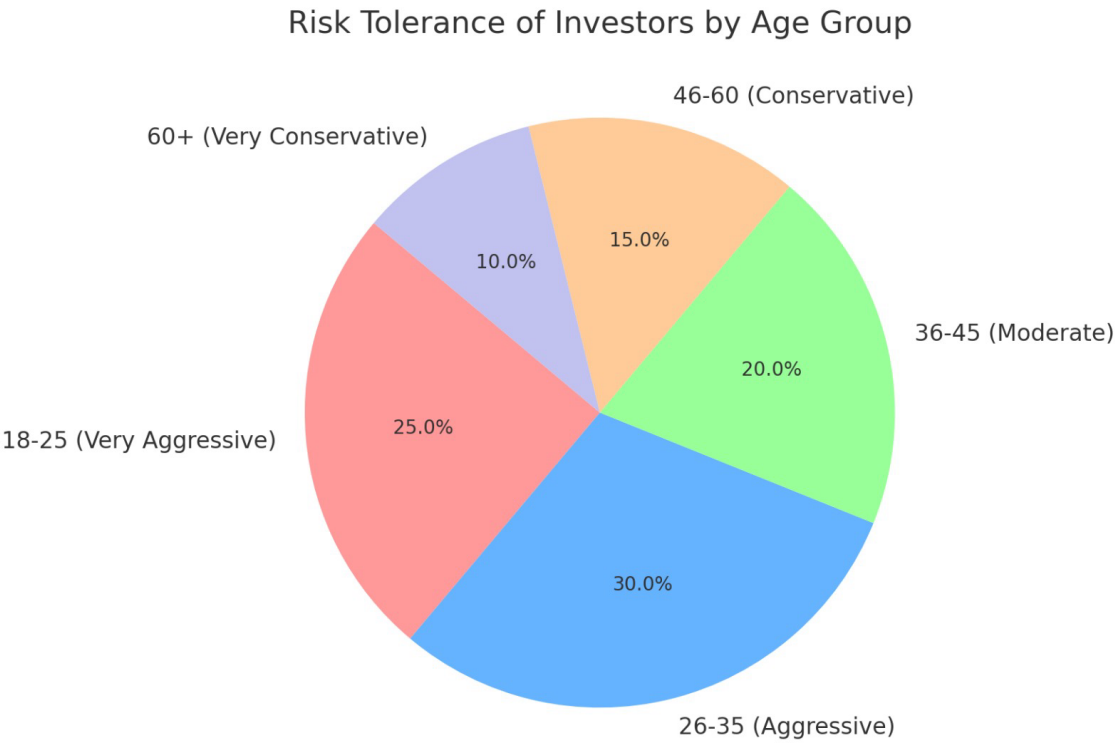


Figure 5: Risk Tolerance of Investors by Age Group

The Proper Asset Allocation Of Stocks & Bonds By Age		
New Life Model		
Age	Stocks	Bonds
0 - 25	100%	0%
30	90%	10%
35	85%	15%
40	80%	20%
45	75%	25%
50	70%	30%
55	65%	35%
60	60%	40%
65	55%	45%
70	50%	50%
75+	45%	55%
Source: FinancialSamurai.com		

Figure 6: The Proper Asset Allocation of Stocks and Bonds by Age

2.4 How the Software Meets Their Needs:

- i. **Common Investors and Beginners:** Provides an easy, guided approach to systematic investing without requiring advanced knowledge.
- ii. **Institutional and Professional Investors:** Enhances decision-making with data-backed stock shortlisting based on HQM analysis. Reduces human biases in mutual fund and hedge fund capital allocation strategies.
- iii. **Passive Investors and Retirees:** Offers a hands-off investment approach to grow wealth systematically over time.
- iv. **Tech-Savvy Enthusiasts:** Serves as a platform to explore algorithmic investing without requiring advanced programming knowledge

2.5 Comparison of Target Audience Needs vs. Solution Features:

User Type	Investment Needs	Solution Features
Retail Investors	Simple, low-risk, automated investment Suggestions	Pre-set investment strategies, risk-adjusted stock selection
Hedge Fund Managers	Data-backed decision-making, capital efficiency	Advanced stock filtering based on HQM analysis

Mutual Fund Managers	Efficient asset allocation, reduced bias	Automated capital weighting recommendations
Private Equity Managers	High-net-worth portfolio optimization	Customizable stock selection and allocation
Passive Investors	Long-term, stable wealth creation	Hands-off, rule-based quantifiable investing

2.6 Competitor Analysis

2.6.1 Identification and Analysis of Competitors Offering Similar Solutions

The **algorithmic investing and portfolio management** space is highly competitive, with several established solutions. Key competitors include:

- i. **Mutual Funds** - A **mutual fund** is a pooled investment vehicle managed by professionals that offers limited flexibility and transparency, as investors have little control over asset choices and insight into the fund's operations.
- ii. **Hedge Funds** - A hedge fund is a privately managed investment fund that offers flexibility in asset selection and strategy but has high entry barriers due to significant capital requirements and fees, while also lacking transparency in operations and risk disclosures.
- iii. **MetaTrader 5** – A popular trading platform with automated trading capabilities but only provide short-term trading strategies..
- iv. **Coinrule**—Advanced automated algorithmic trading for Crypto Currencies

2.7 Assessment of Competitor Strengths, Weaknesses, Opportunities, and Threats (SWOT)

Competitor	Strength	Weakness	Opportunities	Threats
Coinrule	- Advanced trading features - Algorithmic trading support for cryptocurrencies	- Focus on a volatile asset class - Uncertainty around the long-term viability of crypto markets	- Integration of new asset classes (stocks, bonds, etc.) - Broaden appeal by diversifying beyond crypto	- Potential future regulations that could disrupt the entire cryptocurrency market and related business models
Traditional Mutual Funds	- Strong capital backing - Investment in fractional shares - Broad public awareness	- Lack of flexibility - No options for risk or diversification selection - One-size-fits-all approach	- Integration of AI-driven strategies - Ability to offer more personalized solutions to a broader audience	- Risk of becoming obsolete due to lack of transparency and flexibility
MetaTrader 5	- Extensive trading features, algorithmic trading support	- Complex UI, steep learning curve - Doesn't provide longterm investing Strategy	- Integration with new markets	- Competition from modern, low-fee trading apps
Traditional Hedge Funds	- High capital flexibility - Access to a wide range of investment instruments - Exclusive appeal	- High capital entry barriers - High fees - Limited Total Addressable Market (TAM)	- Develop automated solutions to cater to a wider, more specific market - Expand accessibility for a larger audience	- Regulatory changes affecting the brokerage model - Shrinking TAM due to exclusivity

2.8 Feature Comparison Matrix

Attributes	Competitors					
	Financial Literacy App	Coin Rule	Traditional Mutual Fund	Meta Trader 5	Traditional Hedge Funds	
	Flexibility	HIGH	HIGH	LOW	LOW	HIGH
	Risk Appetite Selection	HIGH	LOW	LOW	MEDIUM	HIGH
	Diversification	HIGH	MEDIUM	LOW	MEDIUM	HIGH
	Minimum Capital Required	LOW	LOW	MEDIUM	MEDIUM	VERY HIGH
	Risk of Loss	LOW	VERY HIGH	MEDIUM	HIGH	MEDIUM

Metrics

- LOW
- MEDIUM
- HIGH
- VERY HIGH

2.9 Business Values

2.9.1 Unique Selling Points (USPs) that Set the Proposed Solution Apart

- Data-Driven Stock Selection** – Uses HQM analysis for optimized investment decisions.
- Emotion-Free Investment Strategies** – Removes human bias and improves decision-making accuracy.
- Flexibility in Investment Strategy** – Offers users the ability to tailor their investment approach by adjusting stock selection criteria, capital allocation methods, and portfolio diversification strategies. This adaptability ensures that users can align their investment choices with their specific financial goals, risk tolerances, and market conditions.
- Automated Portfolio Management** – Provides systematic stock selection and allocation based on predefined quantitative models.

2.9.2 Articulation of the Value Proposition for Potential Users

The Software is designed to empower retail investors and small investment firms with data-driven trading strategies, automated stock selection, and advanced market insights. Unlike traditional investment platforms that rely on manual decision-making, this solution uses quantitative models and quantitative analytics to optimize investment performance.

2.9.3 Long-term Benefits:

- i. **Cost Savings:** Reduces dependency on financial advisors by offering systematic stock selection.
- ii. **Improved User Experience:** Simple, intuitive interface catering to both beginner and experienced retail investors.
- iii. **Competitive Advantage:** Provides access to advanced stock selection strategies previously available only to institutional investors.

2.10 Conclusion

The Stock Selection Application presents a cutting-edge solution to the challenges investors face in making informed, data-driven decisions. By leveraging High Quantitative Momentum (HQM) analysis, algorithmic stock selection, and risk-based diversification, the application eliminates human biases and subjectivity, empowering both institutional and retail investors with optimized, rule-based investment strategies.

With a focus on automation, scalability, and precision, the application serves as a vital tool for hedge fund managers, mutual fund managers, private equity professionals, and retail investors alike. It enhances capital allocation efficiency, portfolio diversification, and decision-making accuracy by integrating real-time market insights, adaptive risk assessment, and AI-driven recommendations.

In an era where investors demand transparency, efficiency, and reliability, this solution stands out as a market-driven innovation that transforms the investment process. It simplifies access to quantitative investing models, allowing users to confidently build and manage their portfolios while reducing exposure to emotional biases and speculative trading.

Moving forward, the next steps will involve detailed software development planning, real-world testing, and iterative improvements to ensure seamless integration of the proposed features. With a strong foundation in financial technology and data analytics, this project has the potential to redefine systematic investing and democratize access to professional-grade investment tools.

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